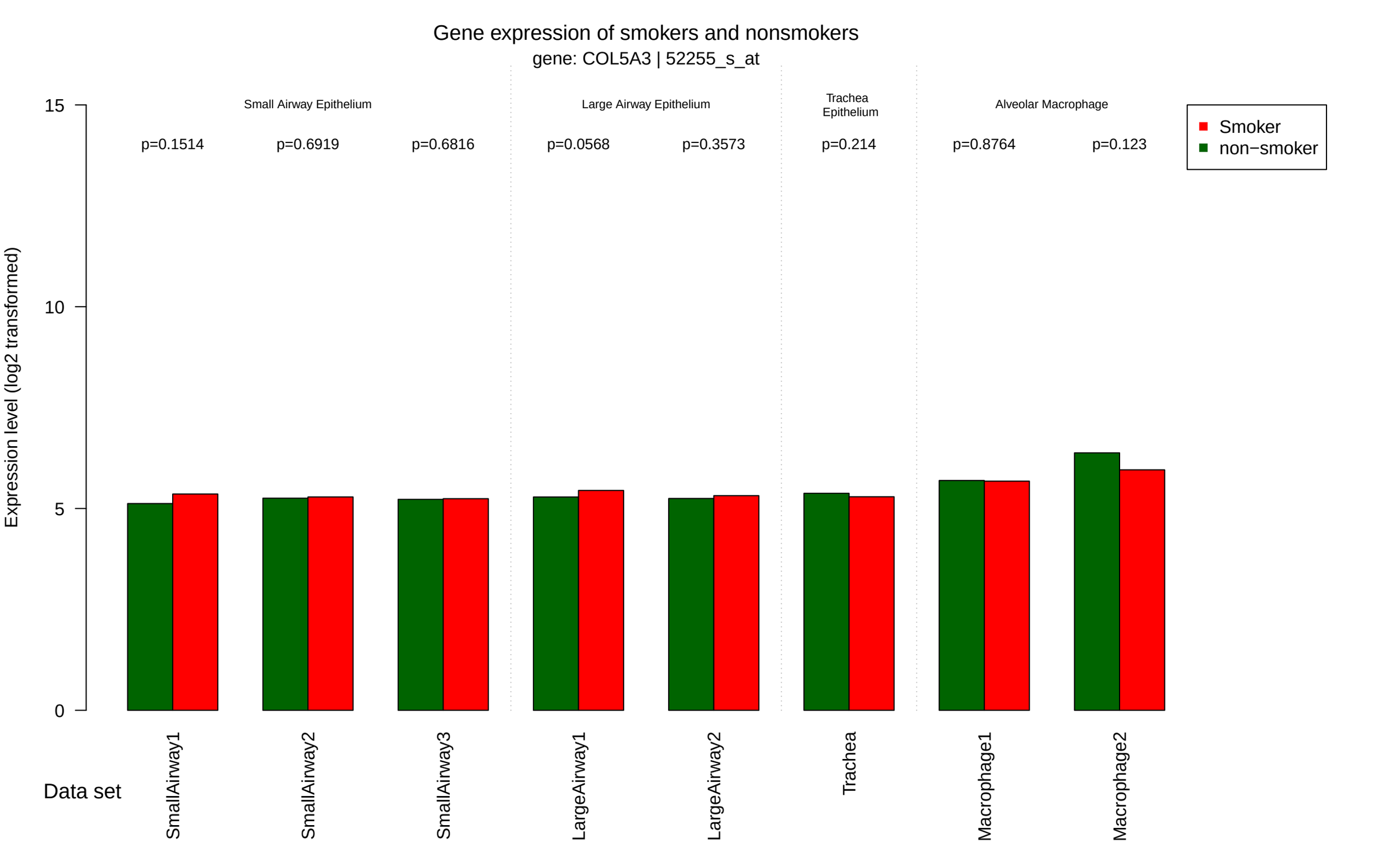
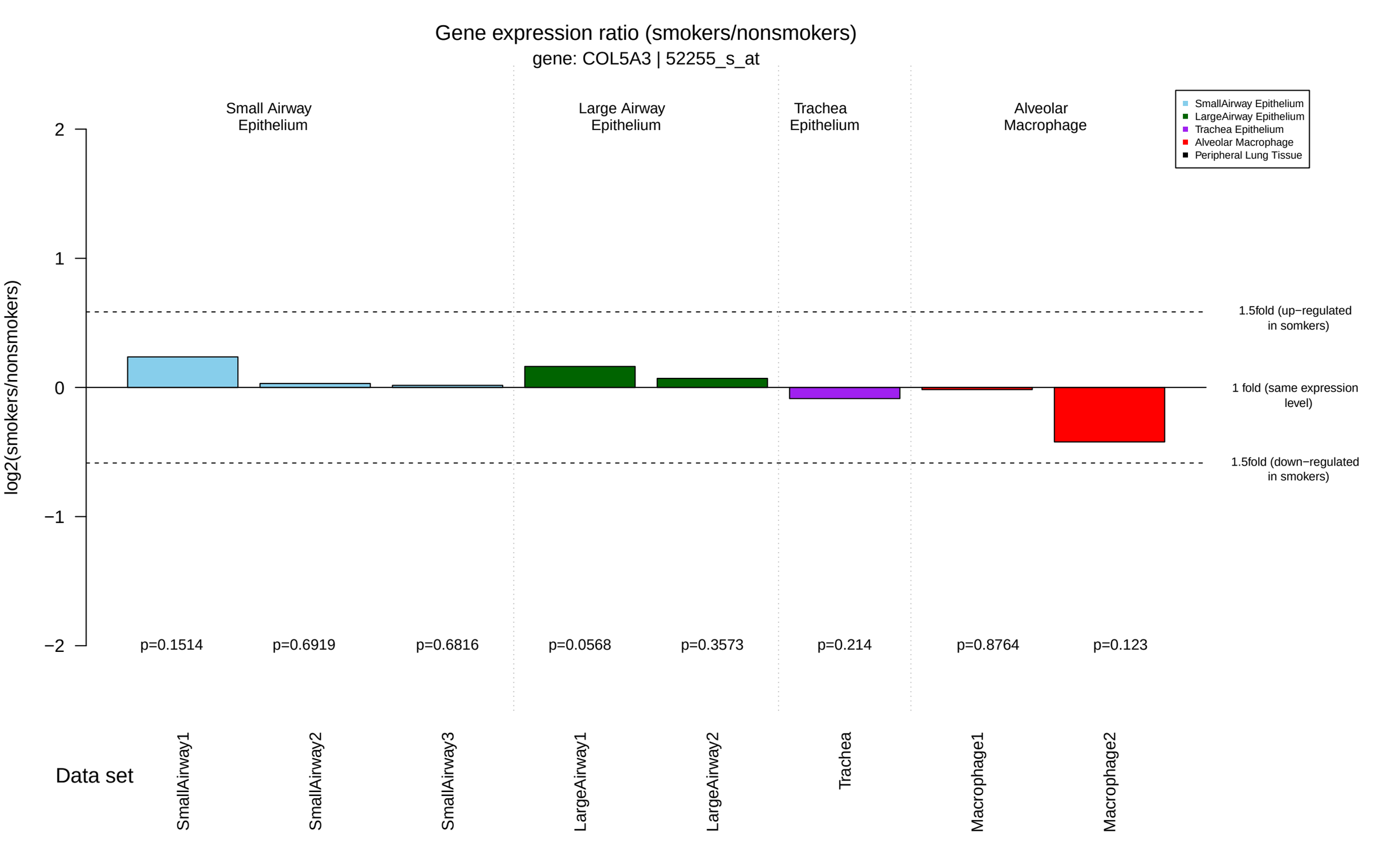
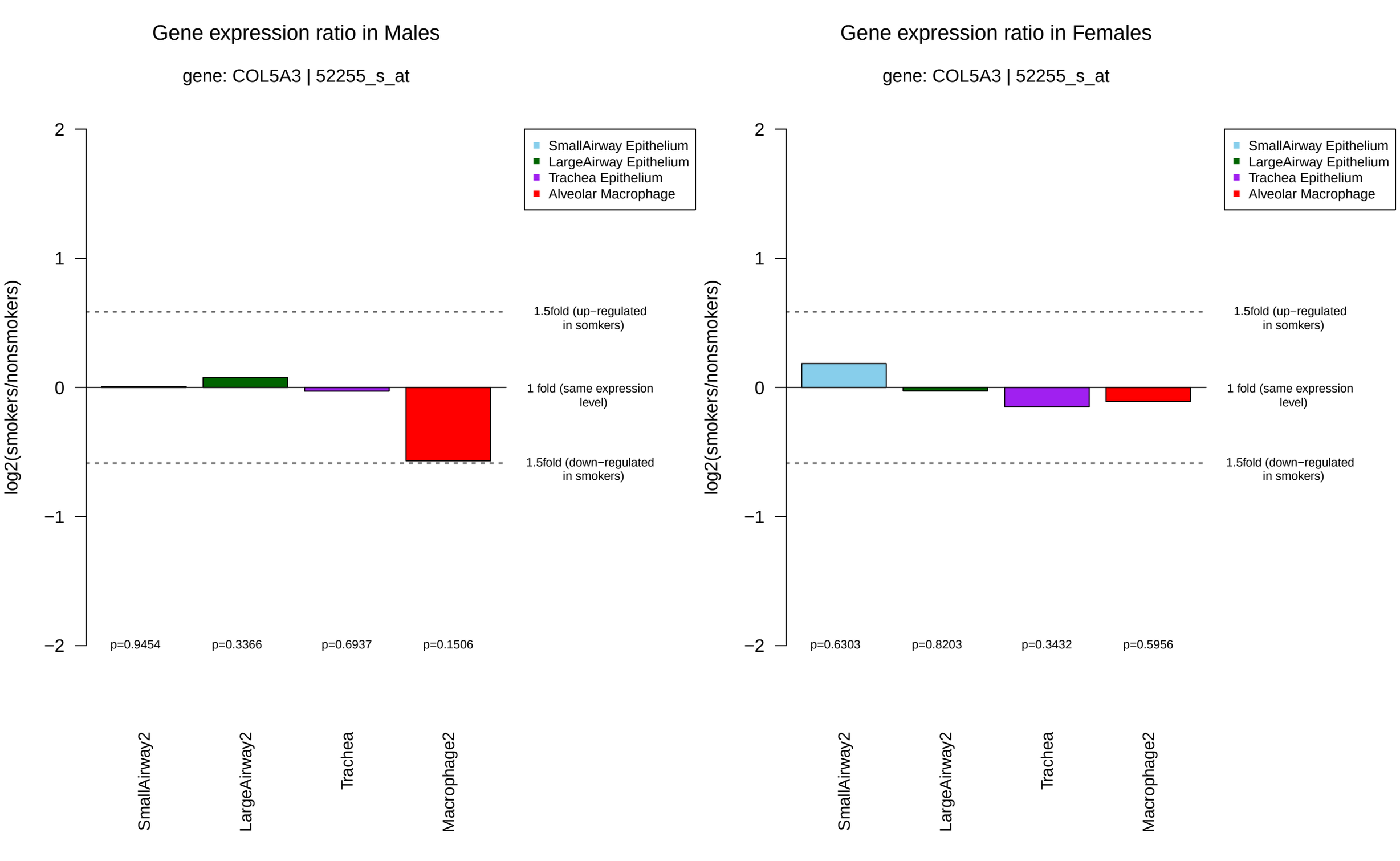
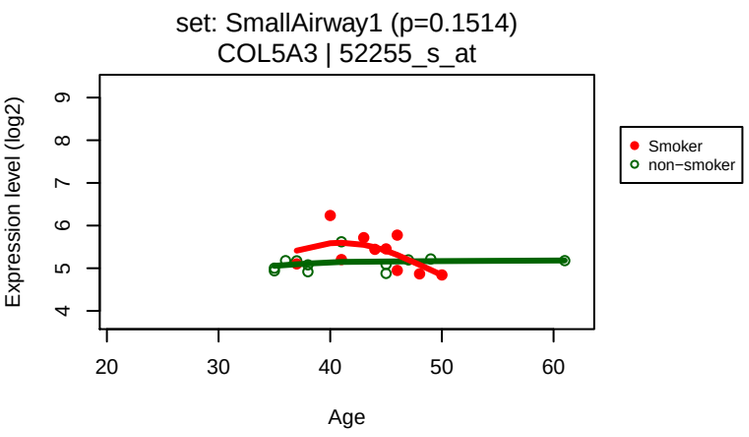
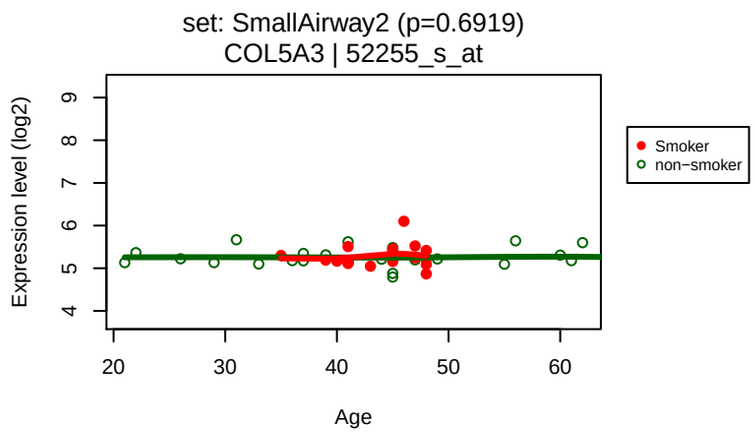
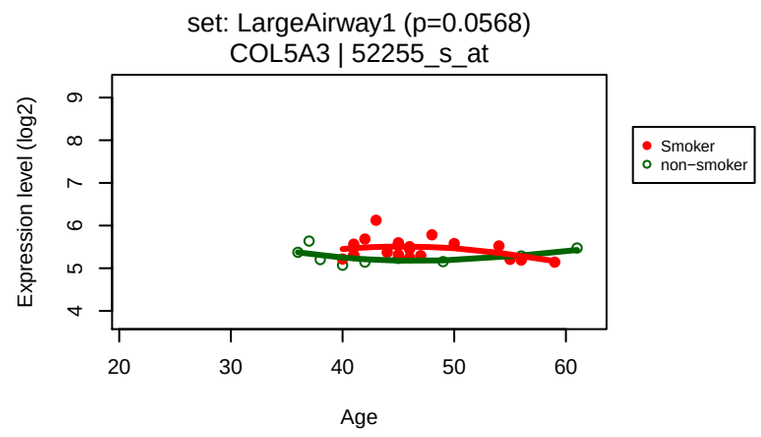

Gene expression of smokers and nonsmokers
gene: COL5A3 | 52255_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1514
p=0.6919
p=0.6816
p=0.0568
p=0.3573
p=0.214
p=0.8764
p=0.123
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

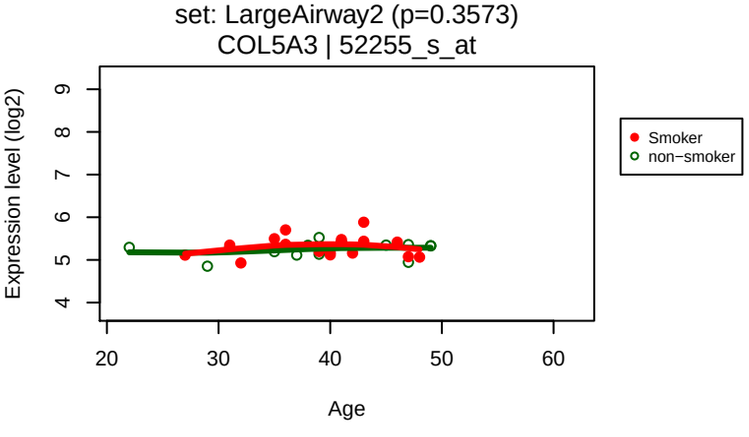

Gene expression ratio (smokers/nonsmokers)
gene: COL5A3 | 52255_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1514
p=0.6919
p=0.6816
p=0.0568
p=0.3573
p=0.214
p=0.8764
p=0.123
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

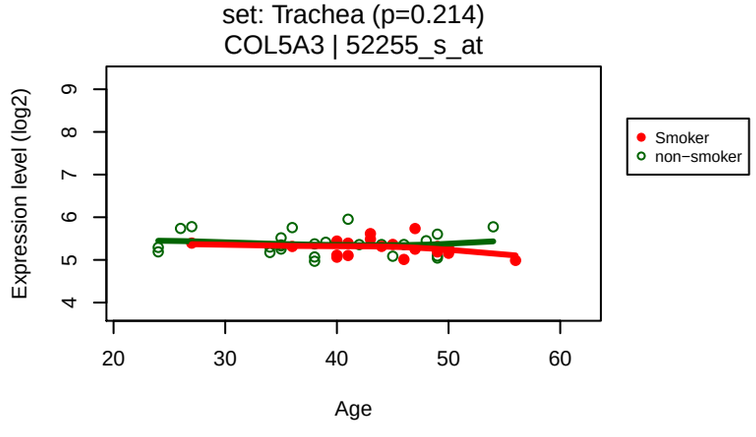

Gene expression ratio in Males
gene: COL5A3 | 52255_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9454
p=0.3366
p=0.6937
p=0.1506
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL5A3 | 52255_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6303
p=0.8203
p=0.3432
p=0.5956
SmallAirway2
LargeAirway2
Trachea
Macrophage2

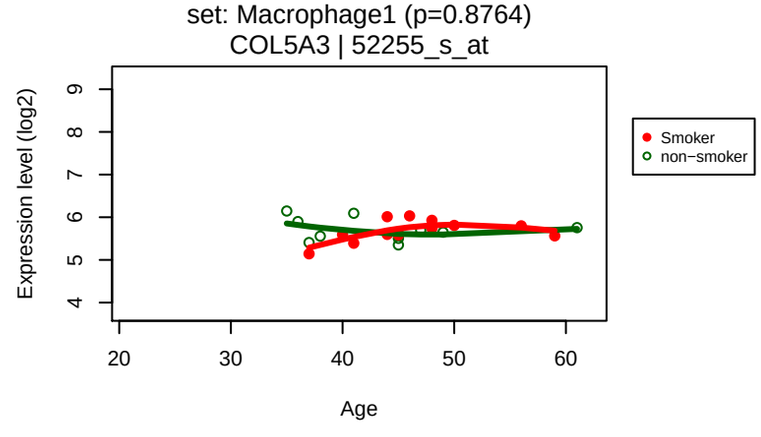

set: SmallAirway1 (p=0.1514)
COL5A3 | 52255_s_at
Expression level (log2)
Age
Smoker
non−smoker

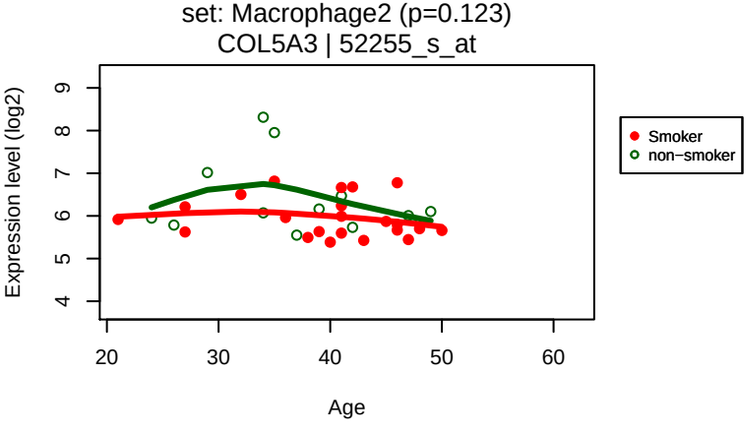

set: SmallAirway2 (p=0.6919)
COL5A3 | 52255_s_at
Expression level (log2)
Age
Smoker
non−smoker

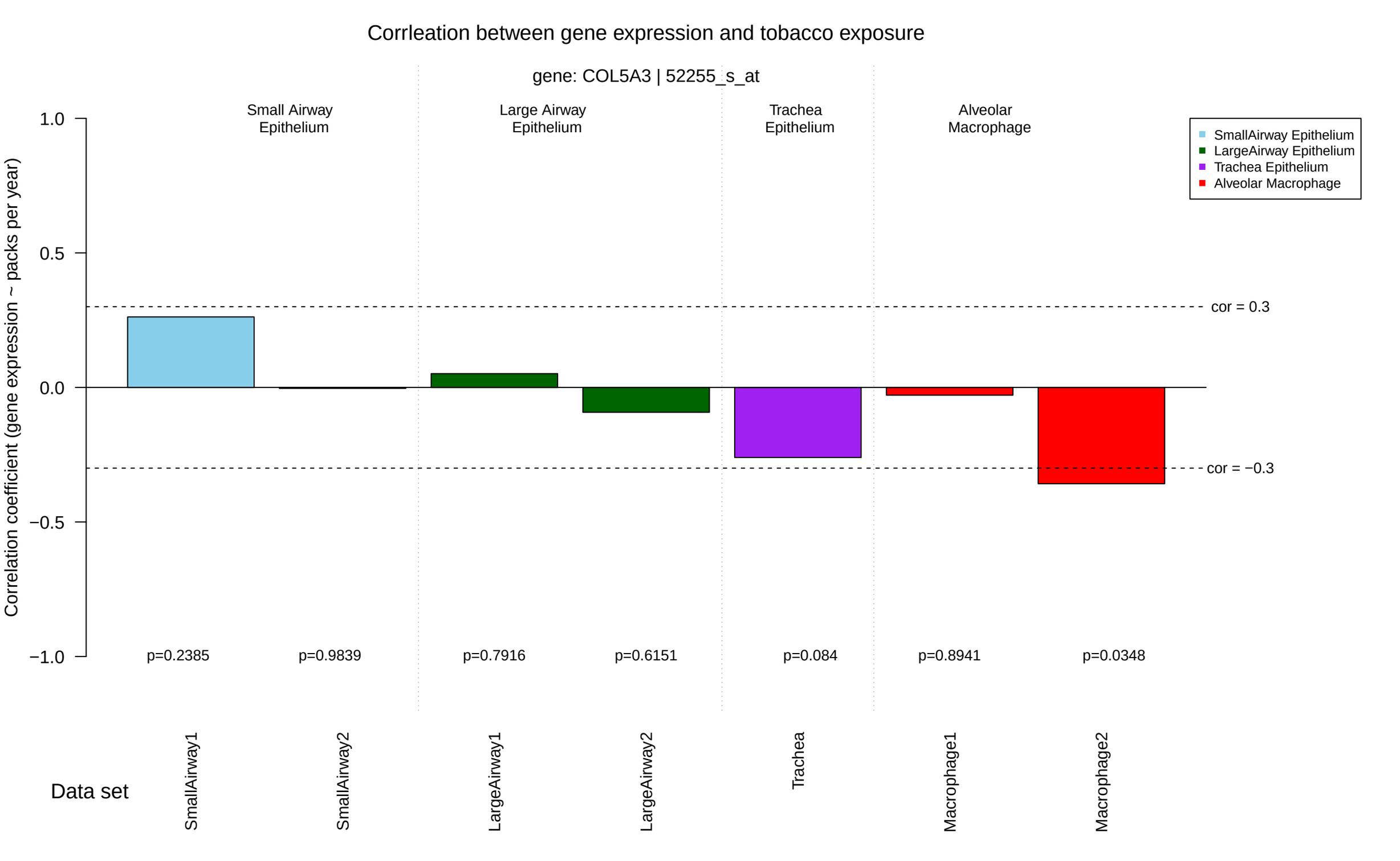

set: LargeAirway1 (p=0.0568)
COL5A3 | 52255_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3573)
COL5A3 | 52255_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.214)
COL5A3 | 52255_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8764)
COL5A3 | 52255_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.123)
COL5A3 | 52255_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL5A3 | 52255_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2385
p=0.9839
p=0.7916
p=0.6151
p=0.084
p=0.8941
p=0.0348
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2